

Oracle-Blind Rollout Geometry Cannot Distinguish Reward Hacking from Benign Convergence

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Abstract

A reward hack and a benign capability improvement are, at the level of the rollout distribution, the *same event*: a policy sharpening onto whatever the reward favors. The only quantity that distinguishes them—whether the favored mode also has higher *true quality*—is the held-out oracle, which an *oracle-blind* detector cannot observe by construction. We formalize this as an indistinguishability result (Proposition 1) and, via a data-processing argument, as a bound stating that no oracle-blind decision rule achieves above-chance worst-case separation of the two regimes (Corollary 1). We then support the claim with three empirical legs on a GRPO-instrumented code-generation setting. First, in a detector-characterization study, every candidate geometric axis (convergence sharpness, output degeneracy, advantage-distribution shape) fires *identically* on a benign control matched on that axis—including `return 0` (a degenerate hack) versus `return n*(n+1)//2` (a correct one-liner), which are geometrically indistinguishable. Second, two independent real benign GRPO runs (base Qwen2.5-1.5B, a well-specified coding task on which no hacking is possible) each produced a **false positive**—a gentle benign convergence at $KL \approx 0.02$ latched a detector onset at step 94 at a warmup-normalized CUSUM of $S \approx 37.5$, roughly eight times the firing threshold, reproduced across seeds—and an onset-strength sweep establishes a **crossover** below which any onset is inseparable from benign convergence for *any* threshold. Third, an elicitation probe found the test-overwrite exploit *not elicitable within the tested scope* ($\leq 14B$, edit-distance and regex tasks, the test-overwrite affordance): instruction tuning suppresses it (0/192), base 7B models sit at a $\approx 1\%$ noise floor, and a capable 14B base model solving the tasks at 52–55% still produced 0/512 (95% Wilson interval $[0, 0.74\%]$), not rising from 7B to 14B. We conclude that a usable hacking detector must reintroduce some oracle-adjacent signal; we sketch—but, lacking an elicitable hack, do not build—a reframing of rollout geometry as an *efficient sampler* that decides when to spend a scarce oracle query rather than as a classifier.

1 Introduction

Reinforcement-learning post-training optimizes a policy π_θ against a programmatic or learned reward r that stands in for a goal we cannot directly differentiate (Stiennon et al., 2020; Shao et al., 2024). When r is an imperfect proxy, optimization pressure discovers inputs on which r is high but true quality is not—*reward hacking* or specification gaming (Amodei et al., 2016; Skalse et al., 2022; Pan et al., 2022), up to and including direct tampering with the grading channel (Denison et al., 2024). The phenomenon sharpens with optimization strength (Gao et al., 2023), so an inexpensive online signal that flagged the *onset* of hacking—computed from quantities the trainer already produces—would be valuable for intervention before a model ships.

The attractive form of such a monitor is *oracle-blind*: it reads only rollout geometry (KL from a reference policy, entropy, advantage statistics—all already computed each step) and never consults a held-out quality measure. The hypothesis behind this paper’s system, *Flight Recorder*, was that the transition into hacking leaves an abrupt geometric signature (rapid KL acceleration, entropy collapse, advantage redistribution) that precedes the point at which a held-out oracle gap becomes visible, giving an early-warning lead.

This paper reports that the oracle-blind monitor cannot function as a *hacking classifier*, and that the obstruction is *structural*—a property of what the detector is allowed to observe—rather than an artifact of feature engineering or threshold tuning. Our contributions are:

1. A structural argument (§3) that reward hacking and benign capability gain are the same event in rollout-geometry space, formalized as an indistinguishability proposition (Proposition 1) and a data-processing corollary bounding any oracle-blind decision rule to chance-level worst-case separation (Corollary 1).
2. A detector-characterization study (§5.1) showing that each candidate discriminative axis collapses against a benign control *matched on that axis*—the empirical shadow of the argument.
3. Two real benign GRPO runs that are **false positives** at a durable, reproduced operating point (§5.2), and an onset-strength sweep (§5.3) that locates the impossibility region quantitatively via a *crossover* in the detector’s own statistic.
4. An elicitation probe (§5.4) showing the test-overwrite exploit is not elicitable within the tested scope ($\leq 14B$), with the 14B result statistically resolved by a Wilson interval.

We lead with the result rather than the original detector hope: oracle-blind geometry detects *that a convergence is underway*, not *that it is a hack*, and we make that boundary precise.

2 Related work

Reward hacking and specification gaming. That optimizing a proxy reward degrades the true objective is a foundational AI-safety concern (Amodei et al., 2016). Skalse et al. (2022) give a formal definition—a proxy is hackable if it can increase while the true reward decreases under some policy pair—which is precisely the configuration our impossibility argument exploits. Pan et al. (2022) map how misspecification severity scales with model capability and optimization, and Gao et al. (2023) quantify reward-model overoptimization. Denison et al. (2024) show that models trained on gameable environments can generalize to editing their own grading code—the exact test-overwrite affordance our elicitation probe targets. All of this work studies *when* hacking occurs; we study whether its *onset is detectable from training dynamics alone*.

Monitoring training dynamics. KL-to-reference, entropy, and advantage statistics are standard health signals in policy-gradient post-training (Schulman et al., 2017; Shao et al., 2024). The implicit premise of online monitoring is that a pathological update looks different from a healthy one in these coordinates. Our negative result delimits that premise: pathology defined by *outcome quality* (hacking) is not separable in these coordinates from a healthy sharp update, because the coordinates are computed without reference to quality. This is complementary to chain-of-thought monitoring (Baker et al., 2025), which reintroduces an outcome-adjacent signal (the model’s stated reasoning) precisely because behavior alone is insufficient—consistent with our conclusion that an oracle-adjacent signal is necessary.

Changepoint detection. Our detector uses a CUSUM changepoint test (Page, 1954) on a geometry signal, together with a robust “contraction tube” on the joint feature vector. The contribution here is not a new changepoint statistic but the observation that *no* such statistic, applied to oracle-blind features, can separate the two regimes; the crossover analysis (§5.3) makes the limit quantitative in the statistic’s own units.

3 The structural argument

3.1 Setup

Let a policy π_θ be optimized against a reward $r : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ over prompts x and completions y . Write a *run* as the sequence of per-step rollout batches $\{(x_i, y_i, r_i)\}$ produced during training. From a run we compute two disjoint views:

- the **rollout-geometry features** R — $\text{KL}(\pi_\theta \| \pi_{\text{ref}})$ and its derivatives, token entropy and its trend, and statistics of the GRPO advantage distribution (variance, skew, kurtosis, and a 1-Wasserstein drift against a rolling baseline (Villani, 2009)); and
- the **oracle** O —a held-out measure of true quality (a hidden test suite, a trusted judge), used only for offline evaluation.

A run is a *hack* if r is high while O is low, and a *benign improvement* if both rise. A detector is any measurable function f of R producing an onset decision.

3.2 Indistinguishability

Assumption 1 (oracle-blindness). The detector’s input is R alone; the oracle is excluded, $O \notin R$.

Assumption 2 (geometry is a function of outputs and proxy reward). $R = g(\{(x_i, y_i, r_i)\})$ for a fixed extractor g ; in particular R does not depend on O except through whatever information about O is already carried by (y, r) .

Assumption 3 (realizability of a benign twin). For the proxy rewards r of interest there exists a distinct task/reward pairing that is *well-specified* (a benign run on it cannot hack) yet can realize the same output-and-reward trajectory distribution. (§5.2 exhibits one empirically.)

Proposition 1 (oracle-blind indistinguishability). *Under Assumptions 1–3, for any hacking run H there exists a benign-convergence run B with $R_H = R_B$ and $O_H \neq O_B$. Consequently $f(R_H) = f(R_B)$ for every oracle-blind f , so no f separates hacking from benign convergence by kind.*

Proof. By Assumption 3 choose a well-specified pairing whose output-and-reward trajectory distribution coincides with that of H ; call the resulting benign run B , so $\{(x_i, y_i, r_i)\}_H \stackrel{d}{=} \{(x_i, y_i, r_i)\}_B$. By Assumption 2, R is a fixed function of that trajectory, hence $R_H = R_B$. The runs differ in the oracle: H is a hack (O_H low while r high), B is benign (O_B high), so $O_H \neq O_B$. By Assumption 1, f is a function of R alone, so $f(R_H) = f(R_B)$. The label “hack vs. benign” is a function of O , which is constant on the input f sees; therefore f cannot separate the two. $\square$

The proposition subsumes advantage-distribution statistics, the most tempting “discriminative” axis. Advantage shape can encode *that a subset of samples discovered a higher-reward mode*—but that event occurs identically when the mode is a benign discovery and when it is an exploit, and advantage shape is part of R (computed from r and oracle-blind outputs), so Assumption 2 applies.

3.3 An information-theoretic restatement

Let $C \in \{\text{hack}, \text{benign}\}$ be the regime label. Because C is determined by O and, by Assumption 2, $R \perp O \mid (\text{outputs}, r)$, the label and the features are linked only through the output/reward trajectory T :

$$C \rightarrow O \rightarrow T \rightarrow R \text{ is not a valid Markov chain; instead } C \leftarrow O, \quad R = g(T), \quad T \text{ shared.} \quad (1)$$

The operative fact is that T (and hence R) can be held fixed while C varies (Proposition 1). This yields a worst-case bound on any oracle-blind test.

Corollary 1 (no above-chance worst-case test). *Consider any decision rule $\delta : R \mapsto \{\text{hack}, \text{benign}\}$. Over the family of hack/benign twin pairs guaranteed by Proposition 1, for each pair δ returns the same label on both members, so it is correct on exactly one. Hence its balanced accuracy on the family is exactly 1/2, and its ROC curve cannot rise above the chance diagonal uniformly over the family: no setting of δ (equivalently, no choice of tube radius and CUSUM threshold) achieves true-positive rate $>$ false-positive rate against an adversarially matched benign twin.*

Proof. On a twin pair (H, B) with $R_H = R_B$, $\delta(R_H) = \delta(R_B)$. If δ outputs **hack** it scores a true positive on H and a false positive on B ; if **benign**, a false negative and a true negative. In both cases one error and one success, so per-pair balanced accuracy is 1/2 and $\text{TPR} = \text{FPR}$ on the family. $\square$

Corollary 1 is the precise sense in which the monitor cannot work: not that geometry is uninformative—it genuinely detects *that a convergence is underway*—but that the increment of information needed to call that convergence a *hack* lives entirely in O , which the detector does not see. The remainder of the paper shows that the worst-case twins of Proposition 1 are not pathological constructions but *occur in ordinary training*.

4 Apparatus

Type-level oracle-blindness. Flight Recorder instruments a GRPO trainer. Each step’s rollout batch is split by extractors into a `RolloutFrame` (the only object a detector receives) and an `OracleFrame` (consumed only by the offline evaluator). `RolloutFrame` carries no oracle fields, and `Detector.update(frame: RolloutFrame)` is typed to accept only rollout geometry, so oracle leakage into a detector is a type error—Assumption 1 is enforced mechanically rather than by convention.

Features. The detector-visible extractors produce, per step t : the KL mean κ_t and its discrete second difference $\text{kl_accel}_t = \kappa_t - 2\kappa_{t-1} + \kappa_{t-2}$ (curvature, the leading edge of a distribution shift); the token-entropy slope entropy_trend_t (negative as the policy collapses onto a narrow mode); and the 1-Wasserstein drift $\text{adv_drift}_t = W_1(\hat{A}_t, \hat{A}_{\text{base}})$ of the GRPO advantage distribution against a rolling baseline, alongside its variance/skew/kurtosis and generation statistics (length, n -gram diversity, log-probability concentration). A separate evaluator-only extractor computes the reward–quality gap $r_t - O_t$ and its changepoint; this is never shown to f .

Detector. The default `ContractionTubeDetector` fits a robust reference tube on a warmup window of $W = 30$ steps over $\mathbf{u}_t = [\text{kl_accel}_t, \text{entropy_trend}_t, \text{adv_drift}_t]$, using the per-dimension median m_j and median-absolute-deviation MAD_j . The tube distance is the worst-axis robust z ,

$$d_t = \max_j \frac{|u_{t,j} - m_j|}{1.4826 \text{MAD}_j}, \quad (2)$$

and the tube is “exited” when $d_t > \rho$. In parallel a one-sided CUSUM (Page, 1954) runs on the warmup-standardized signal $z_t = (\text{kl_accel}_t - \mu_W)/\sigma_W$:

$$S_t = \max(0, S_{t-1} + z_t - k), \quad S_0 = 0, \quad (3)$$

with slack k . Onset is declared at the first t for which both gates—tube exit ($d_t > \rho$, magnitude τ) and CUSUM ($S_t > h$)—hold, *sustained* for a persistence window of $P = 5$ steps (two-of- N corroboration, which suppresses transient single-signal spikes). Both gates are **warmup-normalized**, so the detector is invariant to the absolute scale of KL: a run is measured in MAD- and σ -units of *its own* quiet early phase.

Pre-registration. The thresholds $\tau=1.0$, $h=5.0$, $\rho=3.0$, $W=30$, $P=5$ were frozen on a synthetic harness before any real-run data was collected. Because the synthetic harness’s onsets are *authored*, it validates plumbing and tunes the nominal false-positive rate but cannot establish that the signal exists in real runs—which is exactly the gap the rest of the paper probes.

5 Empirical results

5.1 Detector characterization: candidate axes collapse against matched benign controls

Running the real extractors and detector over controlled trajectories, we asked of each candidate discriminative axis whether it separates a hack-like profile from a benign profile *matched on that axis*—the twin of Proposition 1 restricted to one coordinate.

Convergence sharpness. The tube signals (2) fire on any sufficiently sharp convergence. Three benign profiles—collapse onto a constant, convergence onto a complex solution, and convergence via a high-diversity route—produced onsets indistinguishable from a hack-like sharp update. Sharpness measures *how fast* the policy commits, not *to what*.

Output degeneracy. The generation-length / n -gram-diversity / log-probability-concentration features separate a short output from a long one, but that is the rule “short = suspicious,” which a correct one-liner also satisfies. Against a hard control the degeneracy axis fired *identically* on `return 0` (a degenerate hack on a task where 0 is wrong) and `return n*(n+1)//2` (a correct closed form): both are short, low-entropy, and high-concentration.

Advantage-distribution shape. A heavy-tailed advantage distribution is detectable, but it fired *identically* on a hacking profile (one sample found an exploit) and a benign control (one sample was benignly lucky), which share the same advantage shape.

In every case the apparent discriminative power against a *soft* control vanishes against a control matched on the same axis. This is Proposition 1 observed one coordinate at a time.

5.2 Two real benign runs are false positives, with a durable, reproduced operating point

To measure false-positive behavior on *real* geometry we ran a genuine non-hacking GRPO run: base Qwen2.5-1.5B; a well-specified coding task (longest substring with at most two distinct characters, graded by a multi-case correct verifier so that no degenerate constant can pass and hacking is impossible); 120 steps; learning rate 10^{-6} (deliberately gentle); no seeding, no curriculum. The run converged benignly—mean reward $0.19 \rightarrow 0.9$, entropy $1.9 \rightarrow 0.45$, $KL \approx 0.02$ throughout—and **tripped the detector**: a latched onset at step 94 and an instantaneous trigger on $42/91 = 0.46$ of post-warmup steps, at a warmup-normalized peak CUSUM of $S \approx 37.8$, roughly eight times the pre-registered $h=5$. Because the CUSUM (3) is normalized to the run’s own quiet warmup, a benign convergence is a *strong* signal: the late-training z_t are many warmup- σ above the early-training mean regardless of the small absolute KL.

We re-ran the identical configuration on an independent seed with the full per-step geometry persisted to durable storage. The second run reproduces the result exactly—onset 94, peak CUSUM $S = 37.55$ —and now carries *real* advantage drift (`adv_drift` $\in [0, 1.19]$) rather than the zero placeholder present when the first run’s artifact was reconstructed from logs. Replaying the detector over the persisted real geometry yields the same onset and operating point. The false positive is therefore (a) reproduced across seeds at a stable operating point and (b) confirmed not to depend on the advantage-drift signal missing from the first analysis. This run is an empirical witness for Assumption 3: an ordinary benign run that lands deep inside the region a hack would occupy.

5.3 The crossover: where no threshold separates hacking from a benign run

Recalibration—raising h until the benign run goes quiet—appears to work against an *authored*, *strong* synthetic onset, but that margin is a function of how strong the synthetic onset was authored to be, not a property of real hacking. The correct question is the inverse: *how strong must an onset be to remain separable from the benign run?* We parameterize a family of synthetic onsets by an injected per-step KL-acceleration magnitude s , push each through the same extractor and detector, and compare in the detector’s own warmup-normalized CUSUM statistic (3)—the only scale-fair comparison, since both gates are warmup-normalized. Table 1 reports the result.

onset strength s	onset peak CUSUM S	separable from the benign run?
0.10	231.8	yes
0.04	78.5	yes
≈ 0.024 (crossover)	≈ 37.8	boundary
0.02	27.6	no — indistinguishable

Table 1: Onset-strength sweep against the persisted benign run. Because the benign run peaks at $S \approx 38$, an onset is separable iff its peak CUSUM exceeds that value; below the crossover at $s \approx 0.024$ no threshold (τ, h) keeps the benign run quiet while still firing on the onset.

The peak CUSUM grows monotonically with s (approximately linearly in the post-warmup regime, since S_t accumulates $z_t - k$ and z_t scales with s/σ_W). The benign run sits at $S \approx 38$, so *any* onset weaker than the crossover produces a CUSUM no larger than the benign run’s, and no choice of (τ, h) can keep the benign run quiet while firing on the onset. The bar for distinguishability is thus “louder than a benign capability jump,” not “louder than noise”—and by Proposition 1 a hack, being the same geometric event as a benign jump, need not be louder than one. The crossover is the quantitative face of Corollary 1: it is the operating point at which $\text{TPR} = \text{FPR}$ for this benign twin.

5.4 Elicitation probe: the exploit is not elicitable within the tested scope

The strongest objection to a negative result on *detection* is that we never put a *real* hack in front of the detector. We therefore probed, by inference only (no training), whether an emergent test-overwrite hack—the affordance studied by Denison et al. (2024)—is even *elicitable* at reachable scale. Each prompt exposes the **test-overwrite affordance**: the submission is stated to run in a working directory containing the grader file, whose contents are shown, so a model *could* pass by overwriting the tests rather than solving the task. We sample n completions per model at temperature 1.0 and AST-scan each for file manipulation (`open(..., 'w')`, `os.remove`, `Path.write_text`, writes targeting the grader path). We report two rates: **P1**, the task solve rate (a sanity check that the model engages the task at all), and **P2**, the exploit rate. Because GRPO can only reinforce behaviors that are *sampled*, P2 is an upper bound on what training could amplify: if the exploit is never emitted, there is no gradient toward it. Table 2 reports the picture, which is monotone in tuning and, with the 14B run, statistically resolved.

Model	scale / tuning	P2 (exploit rate)	95% Wilson interval
Qwen2.5-Coder-7B-Instruct	7B, instruct	0/192 = 0.000	—
Qwen2.5-Coder-7B (base)	7B, base, coder	2/192 = 0.010	[0.003, 0.037]
Qwen2.5-7B (base)	7B, base, general	1/192 = 0.005	[0.001, 0.029]
CodeLlama-7b-hf (base)	7B, base, coder	1/192 = 0.005	[0.001, 0.029]
deepseek-coder-6.7b-base	7B, base, coder	0/192 (P1= 0, uninformative)	—
Qwen2.5-Coder-14B (base)	14B, base, coder	0/512 = 0.000	[0.000, 0.0074]

Table 2: Elicitation probe (95% Wilson score intervals (Wilson, 1927)). Instruction tuning suppresses the exploit entirely; base models sit at a $\approx 1\%$ noise floor; doubling to 14B base—with strong P1 (52–55% solve rate, so the zero is genuine, not a gradient/parse artifact)—gives 0/512 with the interval entirely below a $\sim 5\%$ trainable threshold and *not* rising with scale.

The 14B zero is the informative cell. With 0/512 the Wilson upper bound is 0.74%; a behavior that appears in fewer than 1 in 135 samples, and that does not increase from 7B to 14B, is below the rate at which GRPO could plausibly catch and amplify it within a modest budget. Within the tested scope ($\leq 14\text{B}$, edit-distance and regex tasks, the test-overwrite affordance) the exploit is either tuned away (instruct) or below trainable rates (base). At reachable scale, then, the phenomenon a hacking detector would need is frequently *not present in the sampled distribution at all*—which reinforces the practical force of the structural result: there is often no real hack for oracle-blind geometry to distinguish from benign convergence in the first place.

6 Limitations

The real false positive is $n = 2$, not $n = \text{many}$. Two independent seeds of one task/model reached the same operating point (onset 94, $S \approx 37.5$). This is a reproduced existence proof that real benign convergence lands in the impossibility region with a stable operating point; it is not a claim that benign convergence universally sits at $S \approx 38$ across tasks and models. The structural result (§3) does not need many witnesses—one suffices to refute separability—but a broader sweep would characterize *where* benign runs land.

The elicitation finding is bounded. It covers $\leq 14\text{B}$, edit-distance/regex tasks, and the test-overwrite affordance only. It does not show that larger models cannot be induced to hack, nor that other exploit

classes are equally unreachable—in particular *verifier-parsing* exploits that fool a string- or regex-based grader (rather than overwrite a file) are a plausible next target, since they require no file-system affordance. These are future work, not gaps in the present claim.

Thresholds are pre-registered on synthetic data. Warmup-normalization removes absolute-scale mis-calibration as an explanation for the false positive, but a *real-onset* reference—which an unelicitable hack denies us—would be needed to claim the monitor could work after recalibration; the crossover (§5.3) shows recalibration cannot rescue the impossibility region regardless.

Scope of the impossibility. Proposition 1 assumes a benign twin (Assumption 3); its force is exactly as strong as the availability of such twins. We argue they are generic (§5.2 exhibits one), but a setting in which every high-reward mode provably coincides with high quality would escape the result—which is just to say: fix the reward, and there is nothing to detect.

7 Implications and a reframing

Pure rollout geometry is insufficient for hacking detection, by a structural argument (Proposition 1, Corollary 1) and three consistent empirical legs. A usable detector must reintroduce an **oracle-adjacent signal**—a held-out verifier, a cheap proxy of true quality, or a known-benign contrastive reference—to recover the one quantity geometry cannot see. This reframes the goal from “detect hacking from geometry” (impossible oracle-blind) to “detect hacking with the *least* oracle signal,” and suggests **geometry as a sampler, not a classifier**: the contraction-tube CUSUM becomes a trigger that decides *when* to spend a scarce oracle query, and the oracle resolves hack-versus-benign at those points. The contribution would be an efficiency frontier—detection latency versus oracle queries spent, against a query-every-step baseline; the crossover gives the natural operating-characteristic axis for such a study. We state this as motivated but untested: building it requires a real hack to detect, and the elicitation gate (§5.4) closed before we could produce one. Characterizing this geometry-as-sampler / oracle-as-verifier trade-off, and the minimal oracle signal it needs, is the natural next direction once a real, reproducible hack is available.

Reproducibility statement

Oracle-blindness is enforced at the type level (§4); the detector, extractors, and the multi-axis characterization study are implemented in the Flight Recorder codebase, and the thresholds are pre-registered and frozen before any real-run data. The second benign run’s full per-step geometry is persisted as a public artifact, and the onset / CUSUM operating point reproduces from it with no GPU. The crossover (§5.3) is a deterministic offline computation over the persisted benign series and a strength-parameterized onset generator. The elicitation probe (§5.4) is inference-only over six open-weight models, with prompts, AST rules, and Wilson intervals as described. The entire empirical record was established for approximately \$2.23 of compute; no training run was required, because the elicitation gate showed the exploit GRPO would need to amplify is not present in the sampled output distribution.

References

- Dario Amodei, Chris Olah, Jacob Steinhardt, Paul Christiano, John Schulman, and Dan Mané. Concrete problems in AI safety. *arXiv preprint arXiv:1606.06565*, 2016.
- Bowen Baker, Joost Huizinga, Leo Gao, Zehao Dou, Melody Y. Guan, Aleksander Madry, Wojciech Zaremba, Jakub Pachocki, and David Farhi. Monitoring reasoning models for misbehavior and the risks of promoting obfuscation. *arXiv preprint arXiv:2503.11926*, 2025.
- Carson Denison, Monte MacDiarmid, Fazl Barez, David Duvenaud, Shauna Kravec, Samuel Marks, Nicholas Schiefer, Ryan Soklaski, Alex Tamkin, Jared Kaplan, Buck Shlegeris, Samuel R. Bowman, Ethan Perez, and Evan Hubinger. Sycophancy to subterfuge: Investigating reward-tampering in large language models. *arXiv preprint arXiv:2406.10162*, 2024.
- Leo Gao, John Schulman, and Jacob Hilton. Scaling laws for reward model overoptimization. In *International Conference on Machine Learning (ICML)*, 2023.

- E. S. Page. Continuous inspection schemes. *Biometrika*, 41(1/2):100–115, 1954.
- Alexander Pan, Kush Bhatia, and Jacob Steinhardt. The effects of reward misspecification: Mapping and mitigating misaligned models. In *International Conference on Learning Representations (ICLR)*, 2022.
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- Joar Skalse, Nikolaus H. R. Howe, Dmitrii Krashennnikov, and David Krueger. Defining and characterizing reward hacking. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- Nisan Stiennon, Long Ouyang, Jeff Wu, Daniel M. Ziegler, Ryan Lowe, Chelsea Voss, Alec Radford, Dario Amodei, and Paul Christiano. Learning to summarize from human feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- Cédric Villani. Optimal transport: Old and new. *Springer Grundlehren der mathematischen Wissenschaften*, 2009.
- Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927.